

Project Book – HRI

Introduction

The objective of this project was to design and implement an intelligent, autonomous robotic agent capable of visual object identification and responsive navigation. The system was designed to process natural language input to determine a target, perform real-time computer vision analysis to locate the object within a 3D space, and execute movement commands to maintain focus on the target.

System Architecture & Methods

To ensure maintainability and robust error handling, the project employed a Decoupled Modular Architecture. The system is composed of four primary domains, each encapsulated within its own logical unit:

- Perception Module (Vision): Utilizes OpenCV and the Ultralytics YOLO framework for real-time inference.
- Aural Module (Hearing): Implemented via the SpeechRecognition library to convert audio to text tokens.
- Control Module (Movement): Acts as a hardware abstraction layer, translating intent into serial communication protocols.
- Decision Engine (Navigator): The central logic that coordinates inputs to issue instructions to the Control module.

Development and Verification Process

The development followed an Iterative Testing Cycle. We transitioned from standalone module testing—ensuring the YOLO model could detect objects and the navigation logic calculated spatial errors correctly—to integrated main-loop testing within an x86_64 development environment (VS Code). This approach allowed for unit testing and mock hardware validation, ensuring that the algorithmic foundation was sound regardless of the target platform's hardware configuration.

Conclusion

Throughout this project, the primary focus has been the development of a sophisticated, modular software framework. It is important to acknowledge that the final physical implementation—the live demonstration of the robot successfully tracking objects in real-time—was not fully realized within the project's timeframe. Engineering is inherently a process of navigating constraints, and despite the maturity of our software, we encountered significant integration complexities at the intersection of embedded Linux drivers and custom hardware protocols.

We have chosen to submit the high-fidelity, modular codebase developed in our professional development environment as the final project artifact. This code represents the true technical scope and achievement of our work. The logic is mathematically sound, the architecture is robust, and the modules are decoupled and verifiable. This project stands as a complete "Software-Defined Robotics" solution, providing a proven, structured roadmap for any future physical implementation.